
Problem 8.22.

Part a. Let $ADD = \{\langle x, y, z\rangle \mid x, y, z > 0 \text{ are binary integers and } x + y = z\}$.
Show that $ADD \in$ L.

Proof. To show that $ADD \in$ L, we give a log space algorithm M that decides ADD.

$M =$ "On input $\langle x, y, z\rangle$, where x, y and z are binary integers greater than 0:

1. Let i, n_z and m be binary integers on work tape where $n_z = |z|$ and m is maximum of $|x|$ and $|y|$.
2. Let s and c be binary digits on work tape where $s, c = 0$.
3. Repeat for each $i = 1, \ldots, m$:
4. If z does not have i^{th} digit ($i > n_z$), then *reject*.
5. Let s be the sum of c and i^{th} binary digits of x and y. If either x or y does not have i^{th} binary digit, then the digit is 0.
6. If the addition produces a carry, then $c = 1$. Otherwise, $c = 0$.
7. If s and i^{th} digit of z are different, then *reject*.
8. If $c = 1$, then:
9. If z does not contain $(m + 1)^{th}$ digit, then *reject*.
10. If c and $(m + 1)^{th}$ digit of z are different, then *reject*.
11. If z has $(i + 2)^{th}$ digit, then *reject*.
12. Otherwise, if z has $(m + 1)^{th}$ digit, then *reject*.
13. *Accept*."

□

Part b. Let $PAL\text{-}ADD = \{\langle x, y\rangle \mid x, y > 0$ are binary integers where $x + y$ is an integer whose binary representation is a palindrome$\}$. (Note that the binary representation of the sum is assumed not to have leading zeros. A palindrome is a string that equals its reverse.)
Show that $PAL - ADD \in$ L.

Proof. To show that $PAL\text{-}ADD \in$ L, we give a log space algorithm M that decides $PAL\text{-}ADD$. Let

$$BIT = \{\langle x, y, i\rangle \mid x, y, i > 0 \text{ are binary integers, and } i^{th} \text{ digit of } x + y \text{ is } 1\}.$$

The algorithm M uses log space algorithms B that decide BIT.

$M =$ "On input $\langle x, y\rangle$, where x and y are binary integers greater than 0:

1. Let i, j and m be binary integers on work tape.
2. Let m be the maximum of $|x|$ and $|y|$.
3. Run B on input $\langle x, y, (m + 1)\rangle$. If B accepts then $m = m + 1$.
4. Let $i = 1$, and $j = m$.
5. Repeat for m steps:
6. Run B on $\langle x, y, i\rangle$. If B accepts, then $b_i = 1$. Otherwise $b_i = 0$.
7. Run B on $\langle x, y, j\rangle$. If B accepts, then $b_j = 1$. Otherwise $b_j = 0$.
8. If b_i and b_j are different, then *reject*.
9. Increment i by 1, and decrement j by 1.
10. *Accept*."

$B =$"On input $\langle x, y, b\rangle$, where x, y and b are binary integers greater than 0:

1. Let i and m be binary integers on work tape where m is maximum of $|x|$ and $|y|$.
2. Let s and c be binary digits on work tape where $s,\ c = 0$.
3. Repeat for each $i = 1, \ldots, m$:
4. Let s be the sum of c and i^{th} binary digits of x and y. If either x or y does not have i^{th} digit, then the digit is 0.
5. If the addition produces a carry, then $c = 1$. Otherwise, $c = 0$.
6. If $i = b$:
7. If $s = 1$, then *accept*. Otherwise, *reject*.
8. If $c = 1$ and $b = (m + 1)$, then *accept*.
9. *Reject*."

□